

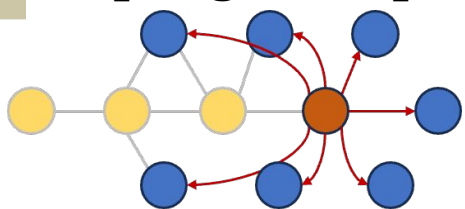
Original Instruction

“Exit the stairs, and then make a right. Walk nearby a sofa and then make a right. Continue walking until you reach game room with a pool table and then wait there.”

Visual Observations



Topological Map



VLM-based Action Router

Temporal Reordering Module

Visual History



Reordered Instruction

1. Exit the stairs.
2. Turn right.
3. Walk nearby a sofa.
4. Turn right.
5. Walk to the game room.
6. Wait at the pool table.

Subgoal Localizer

Current Subgoal

5. Walk to the game room.

Subgoal Reasoning

The last image captures the agent in the living area, already completing walking near the sofa and turning right simultaneously. Therefore, the next logical action is to proceed to the game room.

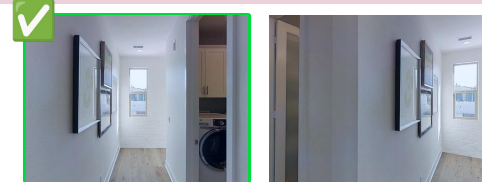
Skill Router

Top-1 Routing

Skill-based Agents

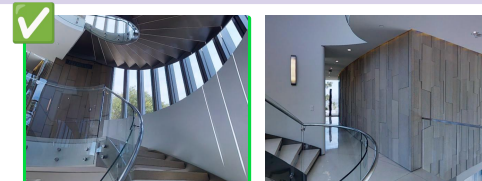
Direction Adjustment

Take a *right turn*.



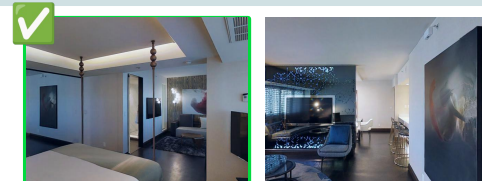
Vertical Movement

Go to the *upper level*.



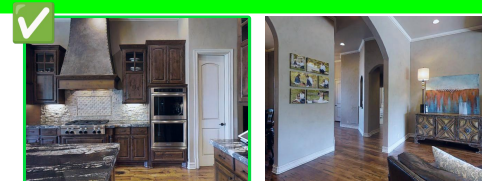
Landmark Detection

Move past the *bed*.



Area and Region Identification

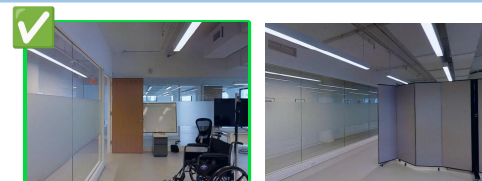
Enter the *kitchen*.



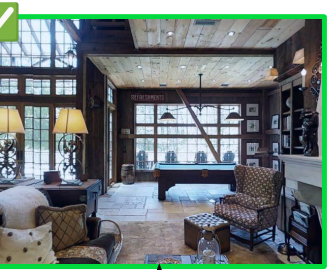
STOP

Stop and Pause

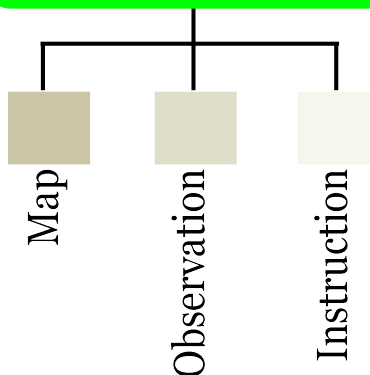
Stop by the *wheelchair*.



Action Image



Area DUET



- Current Location
- Visited Location
- Navigable Location
- 🔥 Fine-tuned Model
- ❄️ Inference Model
- Activated Expert